Jakob Benjamin Wessel

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EDUCATION

PhD in Statistics 09/2022 – 04/2026
University of Exeter, Met Office & The Alan Turing Institute, UK
Topic: *Advances in statistical post-processing, probabilistic forecasting and modelling of extreme events.*
CASE partnership with the UK Met Office. PhD Enrichment Scheme at The Alan Turing Institute (09/2023–07/2024). Research visit to the group of Jakob Zscheischler, Helmholtz Centre for Environmental Research (07/2024–10/2024).

MSc Data Science 09/2020 – 09/2021
Department of Statistical Sciences, University College London (UCL), UK; Final Grade: Distinction; departmental dissertation prize.

BA Philosophy and Political Science 10/2017 – 10/2020
Freie Universität Berlin, Germany; Final Grade: 1.5 (UK: First)
Erasmus exchange: Sorbonne Université Paris, Faculté des Lettres: UFR de philosophie (09/2019–02/2020).

BSc Mathematics and Physics 10/2017 – 10/2019
Technische Universität Berlin, Germany; Final grade: 1.8 (UK: Upper Second)

PROFESSIONAL EXPERIENCE

Institute for Meteorology, University of Leipzig – Postdoctoral Researcher 05/2026 –

- Researcher in the climate attribution group led by Sebastian Sippel.
- Research on AI weather models for understanding and predicting weather extremes, in particular verification for extremes, boosting of AI weather models and rare event algorithms with AI-based sampling guidance.

2 Degrees Investing Initiative, London – Research Analyst 09/2021 – 07/2022

- Developed a methodology to link changes in the climate alignment of financial portfolios to real-economy emission reductions; software implementation in R. Research on climate targets set by financial institutions and evidence on the impact of different strategies on emissions.
- Supported projects analysing financial sector alignment to climate goals in collaboration with finance ministries of Norway and Switzerland.

ACADEMIC ACTIVITIES

Developed the open-source Python package *ibicus*: a flexible toolkit for the bias adjustment of climate models and associated evaluation. The package is used by various organisations within private industry and academia, including by the UK Met Office for providing the local authority climate service. Funded through the ECMWF Code4Earth programme.
PyPI Documentation GitHub

Teaching on the course "Statistical and Machine Learning for Earth System Sciences", University of Leipzig (2026). Developed and held workshops on AI for climate science at Forschungszentrum Jülich (07/2025) and the APARC AI4Climate Summer School.

Co-supervision of PhD student Frieder Loer, together with Sebastian Sippel (primary supervisor) and Thomas Steger, at the University of Leipzig. Topic: *Distributional autoencoders for heatwave attribution.*

SCHOLARSHIPS AND PRIZES

PhD publication prize, University of Exeter, for Wessel et al. (2024), *QJRMS*	04/2025
Helmholtz Visiting Researcher Grant, Helmholtz Information & Data Science Academy	07/2024
Overall Winner, Open Research Awards, University of Exeter, for *ibicus*	08/2023
EPSRC DTP Studentship, University of Exeter	09/2022 – 04/2026
MSc Dissertation Prize, Department of Statistical Sciences, UCL	12/2021
Hackathons: 1st, UCL & Met Office "Hacking the Climate"; 2nd, STUOD, Imperial	2021
Klaus Murmann Scholarship, Stiftung der deutschen Wirtschaft	10/2019 – 09/2021
Deutschlandstipendium (Germany scholarship), Technische Universität Berlin	09/2018 – 09/2019

PUBLICATIONS AS FIRST AUTHOR

Wessel, J. B., Spuler, F., Ferro, C. A. T., Kretschmer, M., Sippel, S. (2026): Probabilistic AI models outperform numerical weather prediction models on record breaking heat extremes. Preprint: DOI.

Wessel, J. B., Allen, S. (2026): Towards Calibrated Probabilistic Forecasts for Events of Interest via Outcome-Conditional Recalibration. Preprint: DOI.

Wessel, J. B., Schillinger, M., Kwasniok, F., Allen, S. (2026): Enforcing tail calibration when training probabilistic forecast models. *International Journal of Forecasting*, 42(4), 1336–1356. DOI.

Wessel, J. B., Chandler, R. (2026): Improving multisite precipitation generators based on generalised linear models. *Adv. Stat. Clim. Meteorol. Oceanogr.*, 12, 149–172. DOI.

Groenke, B.*, **Wessel, J. B.***, Miersch, P., Klein, N., Zscheischler, J. (2026): Stochastic weather generation for scenario-neutral impact assessments using simulation-based inference. *Journal of Geophysical Research: Machine Learning and Computation*, 3, e2025JH000902. DOI.

Spuler, F. R.*, **Wessel, J. B.***, Jebeile, J., Zscheischler, J., Shepherd, T. G. (2026): Bias Adjustment and the Question of Usable Climate Information: Methodological Assumptions and Value Judgments. *Bull. Amer. Meteor. Soc.*, 107, E79–E102. DOI.

Wessel, J. B., Murphy-Barltrop, C. J. R., Simpson, E. (2025): Generative machine learning for multivariate angular simulation. *Extremes* 29, 195–243. DOI.

Wessel, J. B., Ferro, C. A. T., Evans, G. R., Kwasniok, F. (2025): Improving probabilistic forecasts of extreme wind speeds by training statistical post-processing models with weighted scoring rules. *Mon. Wea. Rev.*, 153, 1489–1511. DOI.

Wessel, J. B., Ferro, C. A. T., Kwasniok, F. (2024): Lead-time-continuous statistical postprocessing of ensemble weather forecasts. *Quarterly Journal of the Royal Meteorological Society*, 150(761), 2147–2167. DOI.

Spuler, F. R.*, **Wessel, J. B.***, Comyn-Platt, E., Varndell, J., Cagnazzo, C. (2024): ibicus: a new open-source Python package and comprehensive interface for statistical bias adjustment and evaluation in climate modelling. *Geosci. Model Dev.*, 17, 1249–1269. DOI.

(* joint first authorship)

PUBLICATIONS AS CO-AUTHOR

Burton, C., Di Giuseppe, F., Jones, M. W., Kelley, D. I., [...] **Wessel, J. B.**, and Zheng, B. (2026): State of Wildfires 2025–2026. Accepted at: *Earth Syst. Sci. Data*. Preprint: DOI.

Ribeiro, A. F. S., Billing, M., Thonicke, K., von Bloh, W., **Wessel, J. B.**, Undorf, S., Forkel, M., Zscheischler, J.: Ensemble-based global fire modeling as a tool to characterize extreme wildfire events. Under review at: *Earth System Dynamics*. Preprint: DOI.

Kelley, D. I., Burton, C., Di Giuseppe, F., Jones, M. W., [...] **Wessel, J. B.**, [...], Kolden, C. A. (2025): State of Wildfires 2024–2025. *Earth Syst. Sci. Data*, 17, 5377–5488. DOI.

Jones, M. W., Kelley, D. I., Burton, C. A., Di Giuseppe, [...] **Wessel, J. B.**, [...] Xanthopoulos, G. (2024): State of Wildfires 2023–24. *Earth Syst. Sci. Data*, 16, 3601–3685. DOI.